

# 3GPP TS 33.501

## Security architecture and procedures for the 5G System

Version 17.7.0 Release 17

SYNTHETIC SAMPLE DOCUMENT — written for demonstration and testing of a retrieval-augmented chatbot. This is original content structured to resemble a 3GPP technical specification; it is NOT an official 3GPP publication and must not be used as a source of normative requirements.

### 6.1.3 5G-AKA authentication procedure

5G-AKA is one of two authentication methods standardized for the 5G System, alongside EAP-AKA'. The AUSF requests authentication vectors from the UDM, which derives them using subscriber-specific keys stored in the ARPF. Upon successful completion, both the UE and the network derive a shared anchor key,  $K_{SEAF}$ , from which further session keys are derived. The SEAF forwards the authentication challenge to the UE via the AMF and verifies the UE's response before considering the subscriber authenticated.

### 6.2.1 Key hierarchy

The 5G key hierarchy derives a chain of keys from the long-term subscriber key  $K$ . The Authentication Server Function derives  $K_{AUSF}$ , from which the Security Anchor Function derives  $K_{SEAF}$ .  $K_{SEAF}$  is in turn used by the AMF to derive  $K_{AMF}$ , from which NAS and Access Stratum keys ( $K_{NASint}$ ,  $K_{NASenc}$ ,  $K_{gNB}$ ) are further derived for a given registration.

## 6.2.2 Key derivation summary

The table below summarizes which entity derives each key in the hierarchy and from which parent key it is derived.

| Key    | Derived by | Derived from                 |
|--------|------------|------------------------------|
| K_AUSF | AUSF / UE  | K (long-term subscriber key) |
| K_SEAF | SEAF / UE  | K_AUSF                       |
| K_AMF  | AMF / UE   | K_SEAF                       |
| K_gNB  | AMF / UE   | K_AMF                        |

## 5.9.2 User plane integrity protection

User plane integrity protection between the UE and the gNB is optional to support and to activate, and is subject to a maximum data rate above which integrity protection may be disabled for performance reasons, as configured by the network via the UE security capabilities exchange.